

A Reliability-Constrained Expansion Planning Model for Mesh Distribution Networks

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Abstract—To achieve high reliability, the urban distribution networks are mesh-constructed and radial-operated, in which the outage load can be restored to adjacent feeders via tie-lines after faults. Conventionally, iterative optimization-simulation methods and heuristics are adopted for distribution network planning, which cannot guarantee global optimality. Besides, existing reliability-constrained planning model cannot explicitly assess the reliability indices for mesh distribution networks, so the resulted plan scheme may be overly invested. In this paper, we propose a novel multistage expansion planning model for mesh distribution networks, in which reliability assessment is explicitly implemented as constraints. The different investment/reliability preferences for buses are also customized. Specifically, post-fault load restoration between feeders through tie-lines is modeled as a case of post-fault network reconfiguration. The planning model is then cast as an instance of mixed-integer linear programming and can be effectively solved by off-the-shelf solvers. We use a 54-node system to test the performance of proposed model. Simulation results show the effectiveness and flexibility of this methodology.

Index Terms—Distribution network, multistage expansion planning, reliability assessment, network reconfiguration.

NOMENCLATURE

Indices

a	Index for alternative of equipment.
b	Index for the load levels used to represent the loading condition.
t	Index for stage.
i, j, s, x, y	Indices for nodes.
f	Indices for feeders or corresponding transformers.

Sets

B	Set for the load levels used to represent the loading condition.
Υ	Set of branches.
Λ^C	Set of alternatives of branch.

Λ^T	Set of alternatives of transformer.
Ψ_i	Set of nodes connected to node i .
Ψ^F	Set of feeders.
Ψ_t^{LN}	Set of load nodes at stage t .
Ψ^N	Set of system nodes.
Ψ^{SS}	Set of substation nodes.
Ψ_s^{TR}	Set of transformers belonging to substation at node s .

Parameters

I	Interest rate.
L	Load level.
M	A sufficiently large positive constant.
NC_i^t	Number of customers at node i at stage t .
Δ_b	Duration of load level b .
$C_C^{a,I}$	Investment cost of alternative a for conductors.
$C_C^{a,M}$	Maintenance cost of alternative a for conductors.
$C_S^{s,I}$	Investment cost of substation at node s .
$C_S^{s,M}$	Maintenance cost of substation at node s .
$C_T^{a,I}$	Investment cost of alternative a for transformers.
$C_T^{a,M}$	Maintenance cost of alternative a for transformers.
D_i^t	Peak load demand of node i at stage t .
P_i^t	Active load demand of node i at stage t .
Q_i^t	Reactive load demand of node i at stage t .
S_C^a	Capacity of alternative a for conductors.
T	Planning stages.
T_C^a	Number of stages in the lifespan of the alternative a of conductor.
$T_C^{a,0}$	Number of stages in the remaining lifespan of the existing alternative a of conductor.
T_T^a	Number of stages in the lifespan of the alternative a of transformer.
$T_T^{a,0}$	Remaining lifespan by means of stage for the existing alternative a of transformer.
e_t	Unit vector with $T \times 1$ dimension, whose elements are equal to 0 except the t -element which is equal to 1.
$g_C^{a,\tau}$	Aging vector with $T \times 1$ dimension for conductor alternative a , whose elements are equal to 0 for row $\{1, 2, \dots, \tau - 1\} \cup \{\tau + T_C^a, \dots, T\}$ and are equal to 1 for row $\{\tau, \tau + 1, \dots, \tau + T_C^a - 1\}$.
$g_C^{a,0}$	Aging vector with $T \times 1$ dimension for existing conductor alternative a , whose elements are equal to 0 for row $\{T_C^{a,0} + 1, \dots, T\}$ and are equal to 1 for row $\{1, \dots, T_C^{a,0}\}$.

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$\mathbf{g}_T^{a,\tau}$	Aging vector with $T \times 1$ dimension for transformer alternative a , whose elements are equal to 0 for row $\{1, 2, \dots, \tau - 1\} \cup \{\tau + T_T^a, \dots, T\}$ and are equal to 1 for row $\{\tau, \tau + 1, \dots, \tau + T_T^a - 1\}$.	CIF_i^t	Customer Interruption Frequency at node i at stage t .
$\mathbf{g}_T^{a,0}$	Aging vector with $T \times 1$ dimension for existing transformer alternative a , whose elements are equal to 0 for row $\{T_T^{a,0} + 1, \dots, T\}$ and are equal to 1 for row $\{1, \dots, T_T^{a,0}\}$.	$EENS^t$	Expected Energy Not Supplied at stage t .
$l_{ij}^{a,0}$	Equal to 1 when alternative a for conductors exists at branch ij originally.	$F_i^{f,t}$	$F_i^f \in [0, 1]$ and equals to 1 when node i is supplied by feeder f under normal operating conditions at stage t .
$m_f^{a,0}$	Equal to 1 when alternative a for transformers exists at feeder f originally.	$F_{ij}^{f,t}$	$F_{ij}^f \in [0, 1]$ and equals to 1 when branch ij is supplied by feeder f under normal operating conditions at stage t .
n_s^0	Equal to 1 when a substation exists at node s originally.	$SAIDI^t$	System Average Interruption Duration Index at stage t .
$\varepsilon_i^{CIF,t}$	CIF requirement of node i at stage t .	$SAIFI^t$	System Average Interruption Frequency Index at stage t .
$\varepsilon_i^{CID,t}$	CID requirement of node i at stage t .	$P_i^{xy,t}$	Active demand at node i after post-fault network reconfiguration due to a fault at branch xy (or under normal operation condition if $xy = \text{NO}$) at stage t .
ε_t^{SAIFI}	SAIFI requirement of system at stage t .	$P_{ij}^{xy,t}$	Active power flow through branch ij (from node i to node j) after post-fault network reconfiguration due to a fault at branch xy (or under normal operation condition if $xy = \text{NO}$) at stage t .
ε_t^{SAIDI}	SAIDI requirement of system at stage t .	$Q_i^{xy,t}$	Reactive demand at node i after post-fault network reconfiguration due to a fault at branch xy (or under normal operation condition if $xy = \text{NO}$) at stage t .
ε_t^{EENS}	EENS requirement of system at stage t .	$Q_{ij}^{xy,t}$	Reactive power flow through branch ij (from node i to node j) after post-fault network reconfiguration due to a fault at branch xy (or under normal operation condition if $xy = \text{NO}$) at stage t .
λ_C^a	Outage rate of alternative a for conductors.	$U_i^{xy,t}$	Square voltage at node i after post-fault network reconfiguration due to a fault at branch xy (or under normal operation condition if $xy = \text{NO}$) at stage t .
μ_b	Loading factor of load level b .	$V_i^{xy,t}$	Voltage at node i after post-fault network reconfiguration due to a fault at branch xy (or under normal operation condition if $xy = \text{NO}$) at stage t .
τ_{xy}^{RP}	Duration of the repair-and-switching interruptions associated with the failure of the branch connecting nodes x and y .		
τ_{xy}^{SW}	Duration of the switching-only interruptions that occur when the branch connecting nodes x and y fails.		
ω	Unit reliability cost of EENS ($\omega \geq 0$).		
r_C^a	Resistance unit of alternative a for conductors.		
x_C^a	Inductance unit of alternative a for conductors.		
Binary Variables			
$k_{ij}^{NO,t}$	Equal to 1 when branch ij is connected under normal operating conditions at stage t (which indicates configuration at normal condition).		
$k_{ij}^{xy,t}$	Equal to 1 when branch ij is connected after network reconfiguration due to a fault in branch xy at stage t (which indicates post-fault network reconfiguration strategies).		
$p_i^{xy,t}$	Equal to 1 when node i is affected by the outage due to a fault in branch xy at stage t .		
$q_i^{xy,t}$	Equal to 1 when the demand of node i is completely recovered after reconfiguration of the network following a fault in branch xy at stage t .		
l_{ij}^t	Equal to 1 when there exists a conductor at branch ij at stage t .		
$l_{ij}^{a,t}$	Equal to 1 when alternative a for conductors is installed at branch ij at stage t .		
m_f^t	Equal to 1 when there exists a transformer at feeder f at stage t .		
$m_f^{a,t}$	Equal to 1 when alternative a for transformers is installed at feeder f at stage t .		
n_s^t	Equal to 1 when a substation is newly constructed at node s at stage t .		
Continuous Variables			
CID_i^t	Customer Interruption Duration at node i at stage t .		

I. INTRODUCTION

THE main goal of distribution network planning is to continuously meet the electricity needs of demands with required quantity and quality, which is called reliability [1]. Nowadays, the reliability of a distribution system has been systematically characterized by several indices including customer interruption frequency (CIF), customer interruption duration (CID), the loss of load probability (LOLP), system average interruption frequency index (SAIFI), system average interruption duration index (SAIDI) and the expected energy not supplied (EENS), which are generally taken into account in planning models [2]–[6]. In order to meet the requirements of economic and reliability criteria simultaneously in planning models of distribution networks, the models need to assess the reliability of planned networks during the optimizations.

However, conventionally, these reliability indices are calculated by analytical methods [1], [7]–[14] or simulation-based methods [16]–[20], which are unable to be directly embedded into the planning models of distribution networks. For example, topology analysis is derived in [1], [2], [7] and fault incidence matrices are formulated in [12], [13] based on the topology information of distribution networks, then algebraic equations for reliability indices can be formulated with the failure rates and

the repairing interruptions of equipment. In [14], based on the topology of the distribution network, linear algebraic equations are built and the reliability indices can be linearly expressed by the solution. In [15], based on the fault incidence matrix proposed in [12], an optimal planning model of the sectional switch and tie line for the distribution network is proposed considering the tradeoff between the system reliability and economy. Obviously, these topology-dependent analytical methods need the network's information as a priori, which cannot be implemented into distribution planning models as constraints.

On the other side, for the simulation-based methods, in order to satisfy the reliability requirement of the planned networks, iteration steps including an optimization step and a posterior reliability assessment simulation procedure (called iterative optimization-simulation methods) are often adopted to generate feasible planning solutions [21]–[25]. These solutions could be intuitive and suboptimal for the purpose of 100% guaranteeing the reliability requirement and the ultra-high computational burden would be involved in these methods.

Motivated by the lack of exact methods for reliability-constrained distribution planning problems, recently, several optimization model-based reliability assessment methods have been proposed in [26], [27], which are ready to be integrated into distribution network planning models. In [26], a linear-programming-based approach to reliability assessment is presented, where fictitious power flows are modeled and solved to describe the fault-impact-ranges (FIRs). In work [27], post-fault load restoration through tie-lines are further fully considered in the model-based reliability assessment method. Based on the work in [26], a concrete distribution expansion planning model is presented in [28] where the reliability assessment model proposed in [26] is utilized to capture the reliability indices. In [28], the reliability requirement of the planned network is represented by EENS index, which is further added into the objective function with a weight coefficient. Thus, the economic and reliability criteria are optimized at the same time. Inspired by [26], an MILP model for incorporating reliability indices in distribution system expansion planning is proposed in [29] and further studied in [30] where the economic reward-penalty scheme for system requirement is modeled and the role of distributed generations in reliability assessment is considered.

However, based on the reliability assessment method proposed in [26], which does not take post-fault load restoration strategies between feeders into account, the planning model proposed in [28] may not be suitable for the planning of mesh distribution networks. The planning model presented in [29] and [30] also neglects the operation of normally-open backup switches (tie switches). It has been shown in [27] that such post-fault load restoration strategies can greatly improve the reliability of distribution networks. For the mesh distribution networks, the reliability assessment method proposed in [26] may significantly underestimate its reliability. Thus, the model proposed in [28] will generate overly invested planning scheme to meet specific reliability requirements. Besides, the degradation and ageing models of power equipment are not taken into consideration in [29] and [30], which plays an important role in practical distribution network planning in a long run. For example, in [31] and [32], the degradation model for transformers is

considered into the reliability-centered maintenance scheduling model. And in [33], reliability model for generation adequacy is studied counting the degradation of components.

In urban area, mesh-constructed distribution network configurations are very popular to improve the reliability of power supply [34]–[36]. A mesh-constructed network structure includes redundant tie-lines which are organized as backups for rerouting power in the event of a failure in the main branches, and thus is more reliable than a radial-constructed network [34]. Apart from that, considering post-fault network reconfiguration into planning models of mesh distribution networks have been widely researched in the [37]–[42], proving an evident improvement of system reliability. However, these works still use Monte Carlo simulations and heuristics where a population of candidate topologies is iteratively handled, which cannot obtain global optimality.

Built upon [27], this paper presents a reliability-constrained multistage expansion planning model for mesh distribution networks, where the selection and placement of tie-lines are designed, then network reconfiguration strategies are fully considered and the reliability indices are explicitly counted. Compared to the reliability assessment model presented in [27], in this work, the network topology is not known as a priori, instead being an outcome of the optimization process. Thus, the algebraic expressions of [27] are extended in a nontrivial way, where the existing status of equipment is considered, making a distinctive difference from the original reliability assessment model. Besides, the degradation of components is considered in the proposed model.

Compared to the reliability-constrained distribution network planning model presented in [28], this work steps forward in three aspects. Firstly, the proposed model can cope with mesh distribution networks (where tie-lines need to be planned) and considers post-fault load restoration through tie-lines, which is not included in [28]. Thus, for the mesh-designed distribution networks with post-fault network reconfiguration strategies, the proposed method can explicitly describe reliability indices and satisfy the economic and reliability criteria simultaneously. Secondly, the proposed method can generate best configuration for normal operation of distribution networks (to determine which branches are tie-lines) as the model is solved, and the post-fault network reconfiguration strategies of all considered outages are given as an output of the optimization. Thirdly, differed from [28] where reliability terms are added into objective function, the proposed method restricts the reliability indices more carefully and practically as constraints. Different reliability requirements for different load buses are also handled known as individualized demand importance level [40], to customize various investment/reliability preferences.

The proposed model is finally formulated as a mixed-integer linear program (MILP) and finite convergence to the optimum is guaranteed. Additionally, effective off-the-shelf software based on the branch-and-cut algorithm is readily available [43].

The main contributions of this paper are twofold:

- 1) A reliability-constrained multistage expansion planning model for mesh distribution networks is proposed, in which post-fault load restoration between feeders is fully considered and the reliability indices are explicitly

counted. The proposed model takes MILP formulation and can be effectively solved by off-the-shelf software.

- 2) The different investment/reliability preferences for buses are customized by reliability constraints. The balance between economic and reliability criteria of the planned distribution networks is discussed in case studies.

The remainder of this paper is organized as follows. Section II presents the reliability assessment and planning problem. Section III describes the expansion planning model of distribution networks. In Section IV, numerical tests are reported and analyzed using a 54-node system. Finally, relevant conclusions are drawn in Section V.

II. RELIABILITY ASSESSMENT AND PLANNING PROBLEM

Distribution system reliability is assessed on a yearly basis through well-known indices such as CIF, CID, SAIFI, SAIDI, ASAI, and EENS. CIF and CID account for the frequency and the duration of interruptions at each load node respectively. SAIFI is an index of how many sustained interruptions an average customer will experience. SAIDI is an index of how many interruption hours an average customer will experience. ASAI is the proportion of hours that the service will be available for an average customer. Finally, EENS is defined as the total energy that is expected to be curtailed.

A. Assumptions

To formulate the reliability requirements as constraints in the optimization model, the following practical assumptions are customarily adopted for the sake of tractability [27].

- 1) For simplicity, we assume that the contingency set only contains single branch outages, which are characterized by failure rates and interruption durations. The contingency set is scalable as multiple faults could be added into it if needed.
- 2) We consider the mesh-constructed distribution network operated radially, where each branch is connected to a feeder and each feeder is equipped with a circuit breaker inside the substation. If one end of a branch is not equipped with a switch, then this branch can be merged into the adjoining branch [44]. Thus, we can assume that both ends of each branch are equipped with switches and any faulty branch can be isolated locally. Once a sustained fault occurs, the first circuit breaker upstream of the fault trips, thereby curtailing all downstream load demands. After a switching-only interruption duration (from the fault happens to the network reconfiguration implements), the switches on either side of the breakdown branch are opened to isolate the fault. Then the system topology is reconfigured by operating switches and circuit breakers to recover maximal outage power demand. Finally, once the isolated fault is cleared, after a switching-and-repairing interruption duration (from the fault happens to the damage is repaired), the corresponding switch is closed and the original system topology is reinstated.

A six-bus system is described here as an illustrative example, which includes 2 substation nodes and 4 demand nodes, as

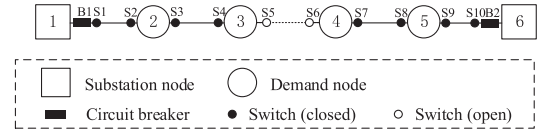


Fig. 1. Illustrative example.

shown in Fig. 1. Specifically, an outage scenario is discussed in detail when a fault happens on the branch 2–3. Just after the fault happens, the fault trips the circuit breaker B1, thus the demands connected to nodes 2, 3 lose power supply. Next, after a switching-only interruption duration of branch 2–3, switches S3 and S4 are opened to isolate and repair the fault, while circuit breaker B1 recloses as well as switches S5 and S6 are closed (which is called post-fault network reconfiguration), so the demands connected to nodes 2, 3 are restored. Then, after a switching-and-repairing interruption duration from the fault happens, the fault is cleared and the original system topology is reinstated. In [26] and [28], the tie-line between nodes 3 and 4 is not modeled thus the load restoration through this tie-line cannot be considered, which underestimates the reliability. Actually, much more complicated load restoration strategies are adopted in reality, which should be incorporated in the model.

B. Planning Problem Descriptions

The proposed multistage expansion planning of mesh-designed distribution network with reliability constraints has been modeled taking the following factors into account [45]:

- The distribution network is composed of nodes at which loads and sources are connected, and branches forming connections between nodes, representing the feeders;
- The planning horizon is divided into T stages of known duration, with the variables in the problem being associated with each stage;
- Possible alterations to network nodes and branches constitute a set of investment alternatives to be used in solving the network expansion problem;
- Capacity expansion of branches and operational voltage constraints are considered in the model; reliability requirements are formulated as explicit constraints considering the post-fault network reconfiguration strategies which include load restoration schemes between feeders.

III. PROPOSED RELIABILITY-CONSTRAINED PLANNING MODEL

In this section, we describe the reliability-constrained multistage expansion planning model for mesh distribution networks based on the reliability assessment model in [27].

A. Objective Function

The objective function for the distribution network expansion planning model includes three parts: the investment cost c_t^I , the reliability cost $\omega EENS^t$ and the maintenance cost c_t^M , and the proposed model is to minimize the present value of the total cost,

which, based on [28] and [45], is formulated as

$$C^{PV} = \sum_{t=1}^T [\delta_t^I c_t^I + \delta_t^O (c_t^M + \omega EENS^t)] \quad (1)$$

where t determines the planning stage. The present value factor for investment and maintenance costs at stage t are

$$\delta_t^I = \frac{1}{(1+I)^{\tau(t)}} \quad (2)$$

$$\delta_t^O = \sum_{\tau=\tau(t)}^{\tau(t+1)-1} \frac{1}{(1+I)^{\tau}} \quad (3)$$

where $\tau(t)$ represent the number of year up to stage t .

The investment cost c_t^I and the maintenance cost c_t^M for stage t are respectively defined as

$$c_t^I = \sum_{ij \in \Upsilon} \sum_{a \in \Lambda^C} C_C^{a,I} l_{ij}^{a,t} + \sum_{f \in \Psi^F} \sum_{a \in \Lambda^T} C_T^{a,I} m_f^{a,t} + \sum_{s \in \Psi^S} C_S^{s,I} n_s^t \quad (4)$$

$$\begin{aligned} c_t^M = & \mathbf{e}_t^T \sum_{ij \in \Upsilon} \sum_{a \in \Lambda^C} C_C^{a,M} \left(\sum_{\tau=1}^t l_{ij}^{a,\tau} \mathbf{g}_C^{a,\tau} + l_{ij}^{a,0} \mathbf{g}_C^{a,0} \right) \\ & + \mathbf{e}_t^T \sum_{f \in \Psi^F} \sum_{a \in \Lambda^T} C_T^{a,M} \left(\sum_{\tau=1}^t m_f^{a,\tau} \mathbf{g}_T^{a,\tau} + m_f^{a,0} \mathbf{g}_T^{a,0} \right) \\ & + \sum_{s \in \Psi^S} C_S^{s,M} \left(\sum_{\tau=1}^t n_s^{\tau} + n_s^0 \right) \end{aligned} \quad (5)$$

In (4)(5), $C_C^{a,I}$, $C_C^{a,M}$ are the investment cost and maintenance cost for alternative branch conductor $a \in \Lambda^C$. $C_T^{a,I}$, $C_T^{a,M}$ are the investment cost and maintenance cost for alternative transformer $a \in \Lambda^T$. And $C_S^{s,I}$, $C_S^{s,M}$ are the investment cost and maintenance cost for substation at node s . Binaries $l_{ij}^{a,t}$, $m_f^{a,t}$ and n_s^t represent at the beginning of stage t , whether to install alternative conductor a at branch ij , whether to install alternative transformer a at feeder f and whether to build new substation at node s respectively. In equation (5), $\mathbf{e}_t^T$ is the transverse of unit vector whose elements are equal to zero except the t -element which is equal to one. Besides, $\mathbf{g}_C^{a,\tau}$, $\mathbf{g}_C^{a,0}$, $\mathbf{g}_T^{a,\tau}$ and $\mathbf{g}_T^{a,0}$ are predefined aging vectors for conductors and transformers, which will be further explained with the availability constraints of equipment in Section III-B.

Specifically, in this paper, the maintenance cost includes periodic (preventive) maintenance cost and corrective maintenance cost. In reality, the reliability of power equipment is influenced by the level of maintenance. The more careful and solid the maintenance is, the higher reliability would the equipment maintain. For simplicity, in the proposed model, the maintenance cost for each equipment is predefined to keep its outage rate as a constant.

B. Constraints in the Problem

The constraints are divided into six blocks. The first block of constraints comes from distflow equations [46] at normal

operation scenario and branch outage scenarios:

$$\sum_{j \in \Psi_i} P_{ij}^{xy,t} + P_i^{xy,t} = 0 \quad \forall i \in \Psi^N \quad (6)$$

$$\sum_{j \in \Psi_i} Q_{ij}^{xy,t} + Q_i^{xy,t} = 0 \quad \forall i \in \Psi^N \quad (7)$$

$$\begin{aligned} -M(1 - k_{ij}^{xy,t}) & \leq U_j^{xy,t} - U_i^{xy,t} + 2(r_{ij}^t P_{ij}^{xy,t} + x_{ij}^t Q_{ij}^{xy,t}) \\ & \leq M(1 - k_{ij}^{xy,t}) \quad \forall j \in \Psi^N \end{aligned} \quad (8)$$

$$U_i^{xy,t} = |V^s|^2 \quad \forall i \in \Psi^F \quad (9)$$

$$\underline{U} \leq U_i^{xy,t} \leq \bar{U} \quad \forall i \in \Psi^N \quad (10)$$

$$-Mk_{ij}^{xy,t} \leq P_{ij}^{xy,t}, Q_{ij}^{xy,t} \leq Mk_{ij}^{xy,t} \quad \forall ij \in \Upsilon \quad (11)$$

$$-S_{ij}^{C,t} \leq P_{ij}^{xy,t}, Q_{ij}^{xy,t} \leq S_{ij}^{C,t} \quad \forall ij \in \Upsilon \quad (12)$$

$$-\sqrt{2}S_{ij}^{C,t} \leq P_{ij}^{xy,t} \pm Q_{ij}^{xy,t} \leq \sqrt{2}S_{ij}^{C,t} \quad \forall ij \in \Upsilon \quad (13)$$

$$P_f^{xy,t} = P_{trf}^{xy,t} \quad \forall f \in \Psi^F, trf \in \Upsilon \quad (14)$$

$$Q_f^{xy,t} = Q_{trf}^{xy,t} \quad \forall f \in \Psi^F, trf \in \Upsilon \quad (15)$$

$$P_f^{xy,t}, Q_f^{xy,t} \leq S_f^{C,t} \quad \forall f \in \Psi^F \quad (16)$$

$$P_f^{xy,t} \pm Q_f^{xy,t} \leq \sqrt{2}S_f^{C,t} \quad \forall f \in \Psi^F \quad (17)$$

$$k_{ij}^{xy,t} \leq l_{ij}^t \quad \forall ij \in \Upsilon \quad (18)$$

$$\forall xy \in \Upsilon \cup \{NO\} \text{ for (6) – (18)}$$

Here, we use superscript xy to describe different scenarios. For the case when $xy = NO$, it indicates the corresponding variables are the ones at normal operation scenario. For the case when $\forall xy \in \Upsilon$, it represents the outage scenario due to a fault in branch xy . Constraints (6)–(9) are power flow constraints where the terms of losses are ignored [46], [47]. $k_{ij}^{xy,t}$ represents whether branch ij is connected after post-fault network reconfiguration due to a fault at branch xy (or under normal operation condition if $xy = NO$) at stage t . And M is a big enough number. When $k_{ij}^{xy,t} = 1$ (the branch ij is connected), the equation (8) is transformed to the simplified branch flow equation. When $k_{ij}^{xy,t} = 0$ (the branch ij is disconnected), the constraint (8) is invalid. The accuracy of the linearized distflow model is further discussed in Appendix A. Constraint (10) is nodal voltage constraint. The linking status of branch ij on scenario xy at stage t is determined by binary variable $k_{ij}^{xy,t}$, and the power flow through branch ij is restricted by constraint (11). Equations (12), (13) are capacity available constraints of branch conductor linearized from quadratic capacity limit [47]–[50], and (16), (17) are capacity constraints of transformers. Constraint (18) imposes the linking status $k_{ij}^{xy,t}$ equal to zero when there is no conductor linking branch ij at stage t .

The second block of constraints comes from nodal recognition of out of service by branch outages and load restoration [27]. Here we define two key binary variables for nodes, $p_i^{xy,t}$ and $q_i^{xy,t}$, which indicate whether the demand at node i is affected

by the outage of branch xy and supplied after post-fault network restoration at stage t . The demand at node i is affected by the outage of branch xy at stage t if and only if $p_i^{xy,t} = 1$. After post-fault network reconfiguration, if and only if $q_i^{xy,t} = 1$, the demand at node i can reach power supply.

$$k_{xy}^{xy,t} = 0 \quad (19)$$

$$F_i^{f,t} + F_{xy}^{f,t} - 1 \leq p_i^{xy,t} \quad \forall f \in \Psi^F \quad \forall i \in \Psi_t^{LN} \quad (20)$$

$$1 - p_i^{xy,t} \leq q_i^{xy,t} \quad \forall i \in \Psi_t^{LN} \quad (21)$$

$$P_i^{xy,t} = P_i^t q_i^{xy,t} \quad \forall i \in \Psi_t^{LN} \quad (22)$$

$$Q_i^{xy,t} = Q_i^t q_i^{xy,t} \quad \forall i \in \Psi_t^{LN} \quad (23)$$

$$\sum_{ij \in \Upsilon} k_{ij}^{xy,t} = \sum_{i \in \Psi_t^{LN}} q_i^{xy,t} \quad (24)$$

$$\forall xy \in \Upsilon \text{ for (19)–(24)}$$

Constraint (19) indicates that branch xy is outage and isolated in current scenario xy . Nodes which share the same feeder affiliation variable $F_i^{f,t}$ with the one of the outage branch xy , $F_{xy}^{f,t}$ (which means that under normal operating conditions at stage t , these nodes and branch xy belong to the same feeder) will be affected by the outage due to the tripping of the upstream circuit breaker, as shown in constraint (20). The feeder affiliation variables will be further explained afterwards in the third block of constraints. After circuit breaker tripping and a switching-only interruption duration τ_{xy}^{SW} , the switches on either side of the breakdown branch xy are opened to isolate the fault. Then the system topology is reconfigured by operating switches and circuit breakers to recover maximal outage power demand, as shown in constraints (21)–(24). For the nodes which are not affected by the outage, the demands of these nodes should not be influenced after post-fault network reconfiguration, which is constrained by (21) (thus the situation of $p_i^{xy,t} = 0$ and $q_i^{xy,t} = 0$ is avoided). In this paper, we assume that it can only resupply all ($q_i^{xy,t} = 1$) or none ($q_i^{xy,t} = 0$) of demand following reconfiguration, written by (22) and (23). Radial operation constraint for the distribution network after post-fault network reconfiguration is demonstrated as (24).

The third block of constraints comes from affiliations to feeders of nodes and branches under normal operating condition. Referring to the protection procedures illustrated in Section II, nodes on the same feeder with the outage branch would be affected by the outage. For example, if a sustained fault happens at branch xy which belongs to feeder f , the nodes supplied by feeder f must be affected and the corresponding $p_i^{xy,t}, \forall i \in \Psi^f$ are equal to one. Therefore, we utilize affiliation variables $F_i^{f,t}, F_{ij}^{f,t}$ to indicate whether node i and branch ij belongs to feeder f under normal operating conditions at stage t respectively. These variables are continuously defined but only take binary values, as shown in constraints (27)–(34).

$$P_i^{NO,t} = P_i^t \quad \forall i \in \Psi_t^{LN} \quad (25)$$

$$Q_i^{NO,t} = Q_i^t \quad \forall i \in \Psi_t^{LN} \quad (26)$$

$$\begin{aligned} & - (1 - k_{ij}^{NO,t})M + F_i^{f,t} \leq F_{ij}^{f,t} \leq (1 - k_{ij}^{NO,t})M \\ & + F_i^{f,t} \quad \forall ij \in \Upsilon \quad \forall f \in \Psi^F \end{aligned} \quad (27)$$

$$\begin{aligned} & - (1 - k_{ij}^{NO,t})M + F_j^{f,t} \leq F_{ij}^{f,t} \leq (1 - k_{ij}^{NO,t})M \\ & + F_j^{f,t} \quad \forall ij \in \Upsilon \quad \forall f \in \Psi^F \end{aligned} \quad (28)$$

$$F_{tr}^{f,t} = k_{tr}^{NO,t} \quad \forall f \in \Psi^F, tr^f \in \Upsilon \quad (29)$$

$$F_{ij}^{f,t} \leq k_{ij}^{NO,t} \quad \forall ij \in \Upsilon \quad \forall f \in \Psi^F \quad (30)$$

$$\sum_f F_i^{f,t} \leq 1 \quad \forall i \in \Psi^N \quad (31)$$

$$\sum_f F_{ij}^{f,t} \leq 1 \quad \forall ij \in \Upsilon \quad (32)$$

$$0 \leq F_i^{f,t} \leq 1 \quad \forall i \in \Psi^N \quad \forall f \in \Psi^F \quad (33)$$

$$0 \leq F_{ij}^{f,t} \leq 1 \quad \forall ij \in \Upsilon \quad \forall f \in \Psi^F \quad (34)$$

Constraints (25), (26) shows the demands of nodes under normal operating conditions. Fictitious flows of affiliation variables are defined by constraints (27)–(30), where constraint (29) represents the source of the fictitious flows and constraints (27)–(28) show that the fictitious flows would pass through branch ij if it is linked under normal operating conditions ($k_{ij}^{NO,t} = 1$). A node or a branch can only be supplied by one feeder, which formulated as (31), (32).

The fourth block of constraints comes from availability of equipment [28]. Binary variables l_{ij}^t and m_f^t represent the existing status of branch ij and transformer f respectively. Constraints (35)–(38) describe relations between existing status (l_{ij}^t), capacity ($S_{ij}^{C,t}$), resistance (r_{ij}^t), inductance (x_{ij}^t) and failure rate (λ_{xy}^t) of branch ij with alternatives of conductor respectively. Constraints (40), (41) show the existing status and capacity of transformer f respectively. Logic constraints between the installation of transformers and the existence of substation are demonstrated as constraints (42), (43).

$$l_{ij}^t = \mathbf{e}_t^T \sum_{a \in \Lambda^C} \left(\sum_{\tau=1}^t l_{ij}^{a,\tau} \mathbf{g}_C^{a,\tau} + l_{ij}^{a,0} \mathbf{g}_C^{a,0} \right) \quad \forall ij \in \Upsilon \quad (35)$$

$$S_{ij}^{C,t} = \mathbf{e}_t^T \sum_{a \in \Lambda^C} S_C^a \left(\sum_{\tau=1}^t l_{ij}^{a,\tau} \mathbf{g}_C^{a,\tau} + l_{ij}^{a,0} \mathbf{g}_C^{a,0} \right) \quad \forall ij \in \Upsilon \quad (36)$$

$$r_{ij}^t = d_{ij} \mathbf{e}_t^T \sum_{a \in \Lambda^C} r_C^a \left(\sum_{\tau=1}^t l_{ij}^{a,\tau} \mathbf{g}_C^{a,\tau} + l_{ij}^{a,0} \mathbf{g}_C^{a,0} \right) \quad \forall ij \in \Upsilon \quad (37)$$

$$x_{ij}^t = d_{ij} \mathbf{e}_t^T \sum_{a \in \Lambda^C} x_C^a \left(\sum_{\tau=1}^t l_{ij}^{a,\tau} \mathbf{g}_C^{a,\tau} + l_{ij}^{a,0} \mathbf{g}_C^{a,0} \right) \quad \forall ij \in \Upsilon \quad (38)$$

$$\lambda_{xy}^t = d_{xy} \mathbf{e}_t^T \sum_{a \in \Lambda^C} \lambda_C^a \left(\sum_{\tau=1}^t l_{xy}^{a,\tau} \mathbf{g}_C^{a,\tau} + l_{xy}^{a,0} \mathbf{g}_C^{a,0} \right) \quad (39)$$

$$m_f^t = \mathbf{e}_t^T \sum_{a \in \Lambda^T} \left(\sum_{\tau=1}^t m_f^{a,\tau} \mathbf{g}_T^{a,\tau} + m_f^{a,0} \mathbf{g}_T^{a,0} \right) \quad \forall f \in \Psi^F \quad (40)$$

$$S_f^{C,t} = \mathbf{e}_t^T \sum_{a \in \Lambda^T} S_T^a \left(\sum_{\tau=1}^t m_f^{a,\tau} \mathbf{g}_T^{a,\tau} + m_f^{a,0} \mathbf{g}_T^{a,0} \right) \quad \forall f \in \Psi^F \quad (41)$$

$$\sum_{\tau=1}^t \sum_{f \in \Psi_s^{TR}} m_f^\tau \leq M \left(\sum_{\tau=1}^t n_s^\tau + n_s^0 \right) \quad \forall s \in \Psi^{SS} \quad (42)$$

$$\sum_{\tau=1}^t n_s^\tau + n_s^0 \leq 1 \quad \forall s \in \Psi^{SS} \quad (43)$$

In order to consider the degradation of conductors and transformers, aging vectors, $\mathbf{g}_C^{a,t}$, $\mathbf{g}_C^{a,0}$, $\mathbf{g}_T^{a,t}$ and $\mathbf{g}_T^{a,0}$, are defined as

$$\mathbf{g}_C^{a,\tau} = [0 \cdots 0 \underset{\tau}{1} \cdots \underset{\tau+T_C^a-1}{1} 0 \cdots 0]^T \quad (44)$$

$$\mathbf{g}_C^{a,0} = [1 \cdots 1 \underset{T_C^{a,0}}{0} \cdots 0]^T \quad (45)$$

$$\mathbf{g}_T^{a,\tau} = [0 \cdots 0 \underset{\tau}{1} \cdots \underset{\tau+T_T^a-1}{1} 0 \cdots 0]^T \quad (46)$$

$$\mathbf{g}_T^{a,0} = [1 \cdots 1 \underset{T_T^{a,0}}{0} \cdots 0]^T \quad (47)$$

where T_C^a is the number of stage in the lifespan of the alternative a of conductor, and T_C^0 is the number of stage in the remaining lifespan of the existing alternative a of conductor. T_T^a is the number of stage in the lifespan of the alternative a of transformer, and T_T^0 is the number of stage in the remaining lifespan of the existing alternative a of transformer. Taking constraint (35) as an example, $\sum_{a \in \Lambda^B} (\sum_{\tau=1}^t l_{ij}^{a,\tau} \mathbf{g}_C^{a,\tau} + l_{ij}^{a,0} \mathbf{g}_C^{a,0})$ represents the existence of conductor at branch ij through the whole planning horizon (T stages) as a vector with $T \times 1$ dimension, which is multiplied by unit vector $\mathbf{e}_i^T$ to obtain the existence of conductor at branch ij at stage t .

The fifth block of constraints comes from calculations of reliability indices. Based on the variables p_i^{xy} and q_i^{xy} defined in (20)–(24) [26]–[28], common used reliability indices take linear expressions with these variables, as shown in constraints (48)–(52). For example, CIF of node i traverses the set of outage and counts the time been affected ($p_i^{xy,t} = 1$), as shown in equation (48). For the CID of node i , referring to the action strategies illustrated in Section II, the demand at node i would suffer interruption durations as zero (unaffected with $p_i^{xy,t} = 0$ and $q_i^{xy,t} = 1$), τ_{xy}^{SW} (affected but restored with $p_i^{xy,t} = 1$ and $q_i^{xy,t} = 1$) and τ_{xy}^{RP} (affected and unrestored with $p_i^{xy,t} = 1$ and $q_i^{xy,t} = 0$) when a fault occurs at branch xy , as shown in equation (49) and Fig. 2. The situation of $p_i^{xy,t} = 0$ and $q_i^{xy,t} = 0$ has been avoided by constraint (21). Referring to [4], based on CIFs and

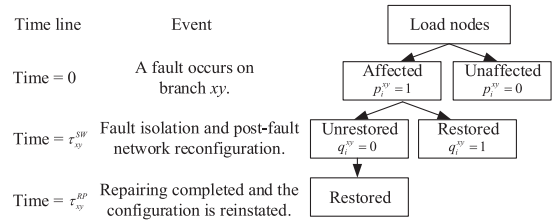


Fig. 2. Illustration of the constraint (49).

CIDs, the systematic reliability indices are expressed as (50)–(52).

$$CIF_i^t = \sum_{xy \in \Upsilon} \lambda_{xy}^t p_i^{xy,t} \quad \forall i \in \Psi_t^{LN} \quad (48)$$

$$CID_i^t = \sum_{xy \in \Upsilon} \lambda_{xy}^t \tau_{xy}^{SW} p_i^{xy,t} + \sum_{xy \in \Upsilon} \lambda_{xy}^t \tau_{xy}^{RP} (1 - q_i^{xy,t}) \quad \forall i \in \Psi_t^{LN} \quad (49)$$

$$SAIFI^t = \sum_{i \in \Psi^{LN}} NC_i^t CIF_i^t / \sum_{i \in \Psi^{LN}} NC_i^t \quad (50)$$

$$SAIDI^t = \sum_{i \in \Psi^{LN}} NC_i^t CID_i^t / \sum_{i \in \Psi^{LN}} NC_i^t \quad (51)$$

$$EENS^t = \sum_{b \in B} \frac{\Delta_b}{8760} \sum_{i \in \Psi_t^{LN}} CID_i^t \mu_b D_i^t \quad (52)$$

The sixth block of constraints are reliability requirements. Different reliability requirements could be applied to different nodes and different stages, as shown in (53)–(57).

$$CIF_i^t \leq \varepsilon_i^{CIF,t} \quad \forall i \in \Psi_t^{LN} \quad (53)$$

$$CID_i^t \leq \varepsilon_i^{CID,t} \quad \forall i \in \Psi_t^{LN} \quad (54)$$

$$SAIFI^t \leq \varepsilon_{SAIFI}^t \quad (55)$$

$$SAIDI^t \leq \varepsilon_{SAIDI}^t \quad (56)$$

$$EENS^t \leq \varepsilon_{EENS}^t \quad (57)$$

The total planning model can be presented as: *minimize* (1)

$$\text{subject to (2)–(57)} \quad \forall t \in \{1, 2, \dots, T\}$$

The proposed model evidently takes mixed-integer linear programming (MILP) formulation and can be effectively solved by off-the-shelf commercial software. The numbers of nodes, branches, transformers and substations in the distribution network are supposed to be n_N , n_B , n_T and n_{SS} , and the most of variables are listed in the following Table I with their dimension. The type numbers of conductors and transformers are known as n_Λ^C and n_Λ^T .

C. Illustrative Application on the Six-bus Network of the Reliability Indices Assessment Model

In order to further explain how the reliability indices are assessed in the planning model, the proposed constraints are applied to the six-bus network in Fig. 1. Specifically, we consider

TABLE I
DECISION VARIABLES AND THEIR DIMENSIONS

Decision variable	Type	Dimension
$\{k_{ij}^{NO,t}\}, \forall ij \in \Upsilon, \forall t$	Boolean	Tn_B
$\{k_{ij}^{xy,t}\}, \forall ij \in \Upsilon, \forall xy \in \Upsilon, \forall t$	Boolean	Tn_B^2
$\{l_{ij}^t\}, \forall ij \in \Upsilon, \forall t$	Boolean	Tn_B
$\{U_{ij}^{a,t}\}, \forall ij \in \Upsilon, \forall a \in \Lambda^C, \forall t$	Boolean	$Tn_B n_\Lambda^C$
$\{m_f^t\}, \forall f \in \Psi^F, \forall t$	Boolean	Tn_T
$\{m_f^{a,t}\}, \forall f \in \Psi^F, \forall a \in \Lambda^T, \forall t$	Boolean	$Tn_T n_\Lambda^T$
$\{n_s^t\}, \forall s \in \Psi^{SS}, \forall t$	Boolean	Tn_{SS}
$\{p_i^{xy,t}\}, \forall i \in \Psi^N, \forall xy \in \Upsilon, \forall t$	Boolean	$Tn_N n_B$
$\{q_i^{xy,t}\}, \forall i \in \Psi^N, \forall xy \in \Upsilon, \forall t$	Boolean	$Tn_N n_B$
$\{p_i^{xy,t}\}, \forall i \in \Psi^N, \forall xy \in \Upsilon, \forall t$	Float	$Tn_N n_B$
$\{Q_i^{xy,t}\}, \forall i \in \Psi^N, \forall xy \in \Upsilon, \forall t$	Float	$Tn_N n_B$
$\{U_i^{xy,t}\}, \forall i \in \Psi^N, \forall xy \in \Upsilon, \forall t$	Float	$Tn_N n_B$
$\{P_{ij}^{xy,t}\}, \forall ij \in \Upsilon, \forall xy \in \Upsilon, \forall t$	Float	Tn_B^2
$\{Q_{ij}^{xy,t}\}, \forall ij \in \Upsilon, \forall xy \in \Upsilon, \forall t$	Float	Tn_B^2
$\{f_i^{f,t}\}, \forall i \in \Psi^N, \forall f \in \Psi^F, \forall t$	Float	$Tn_N n_T$
$\{f_{ij}^{f,t}\}, \forall ij \in \Upsilon, \forall f \in \Psi^F, \forall t$	Float	$Tn_B n_T$

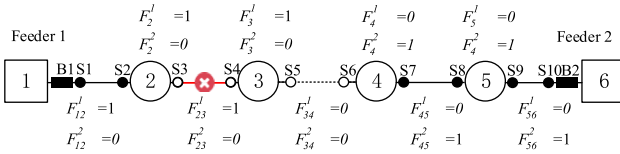


Fig. 3. Values of the affiliation variables for the six-bus system.

the situation when a fault occurs on the branch 2–3. Ignoring the superscript of stage t , at this circumstance, the second block of constraints is described as:

$$k_{23}^{23} = 0 \quad (58)$$

$$F_i^1 + F_{23}^1 - 1 \leq p_i^{23}, i = 2, 3, 4, 5 \quad (59)$$

$$F_i^2 + F_{23}^2 - 1 \leq p_i^{23}, i = 2, 3, 4, 5 \quad (60)$$

$$1 - p_i^{23} \leq q_i^{23}, i = 2, 3, 4, 5 \quad (61)$$

$$P_i^{23} = P_i q_i^{23}, i = 2, 3, 4, 5 \quad (62)$$

$$Q_i^{23} = Q_i q_i^{23}, i = 2, 3, 4, 5 \quad (63)$$

$$k_{12}^{23} + k_{23}^{23} + k_{34}^{23} + k_{45}^{23} + k_{56}^{23} = q_2^{23} + q_3^{23} + q_4^{23} + q_5^{23} \quad (64)$$

As depicted as Fig. 3, under normal operating condition, branch 2–3 is supplied by feeder 1 with $F_{23}^1 = 1$, $F_{23}^2 = 0$. For node 2 and node 3 which are also supplied by feeder 1, restricted by constraint (59), we have $p_2^{23} = 1$ and $p_3^{23} = 1$ which means the demands at nodes 2–3 are affected by the outage on branch xy .

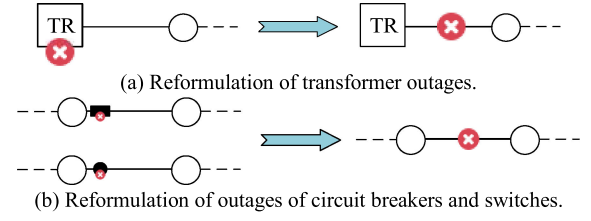


Fig. 4. Reformulation of the outages of transformers, circuit breakers and switches.

On the other side, after post-fault network reconfiguration, q_i^{23} , $i = 2, 3, 4, 5$ are used to indicate whether the demand at node i is supplied after post-fault network reconfiguration. Therefore, by traversing all outage scenarios in the optimization model as constraints, the reliability indices can be expressed by the indicating variables p_i^{xy} and $q_i^{xy,t}$ for $\forall xy \in \Upsilon$.

D. Discussions on the Contingency Set

As illustrated in Section II, the contingency set can be expanded if needed. Several outage scenarios are further discussed here for the integration into the proposed model.

1) *Outages Due to Single Fault*: The branch outage shown as (19) is an example for the outage due to the single fault. Apart from that, single outages of transformers, circuit breakers and switches can also induce interruption of power supply for load demands. On the basis of the proposed model, these outages can be equivalently reformulated as the outages of branches. As shown in Fig. 4(a), the outage of a transformer can be reformulated as the outage of the outgoing branch of the transformer. The equivalent outage rate and switching-and-repairing interruption duration for the branch can be calculated as

$$\lambda'_{trf} = \lambda_{trf} + \lambda_f \quad (65)$$

$$\lambda'_{trf} \tau_{trf}^{RP'} = \lambda_{trf} \tau_{trf}^{RP} + \lambda_f \tau_f^{RP} \quad (66)$$

where λ_f and τ_f^{RP} are the outage rate and repairing duration of transformer f respectively. As shown in Fig. 4(b), the outage of a switching device (a circuit breaker or a switch) can be reformulated as the outage of the belonged branch and the equivalent outage rate and switching-and-repairing interruption duration for the branch can be calculated as

$$\lambda'_{xy} = \lambda_{xy} + \lambda_{xy}^{sd} \quad (67)$$

$$\lambda'_{xy} \tau_{xy}^{RP'} = \lambda_{xy} \tau_{xy}^{RP} + \lambda_{xy}^{sd} \tau_{xy}^{RP, sd} \quad (68)$$

where λ_{xy}^{sd} and $\tau_{xy}^{RP, sd}$ are the outage rate and repairing duration of the switching device at the branch xy .

2) *Outages Due to Multiple Faults*: Outages due to multiple faults may occur with much lower possibilities when several equipment in distribution networks simultaneously breaks down. As illustrated in [27], as a typical multiple fault, the protection failure influence little on the reliability indices of distribution networks (less than 1%). The outages caused by upper transmission/generation systems can lead to multiple fault which power off one or more transformers in the distribution network.

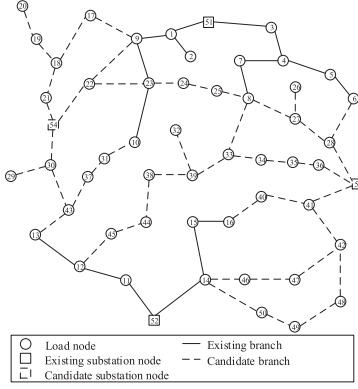


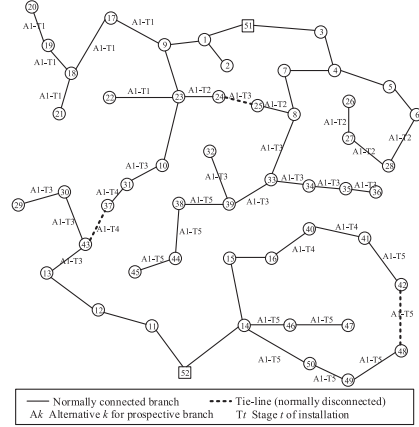
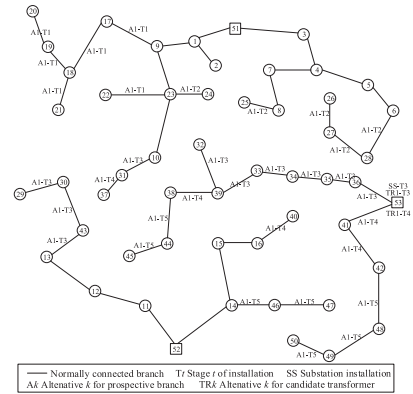
Fig. 5. One-line diagram of 54-node system.

These multiple faults can be added to the contingency set of the proposed model with predefined faulty-event constraints as (19), outage rates and repairing durations. Constraints (6)–(24) are then formulated for these outage scenarios and added to the planning model.

IV. NUMERICAL TESTS

We use a 54-node system described in [42] to test the proposed model. As shown in Fig. 5, the system comprises 50 load nodes, 4 substation nodes, and 63 branches. For reproducibility purpose, system data can be accessed from [51], where base values for power and voltage are 1 MVA and 13.5 kV, respectively, and the currency used is U.S. dollars. The investment and maintenance cost coefficients are based on those provided in [6]. Two types of transformer alternatives and three types of conductor alternatives are considered. Failure rate for feeders is set as 0.1 per year and kilometer [6], and interruption durations for repairing-and-switching and switching-only are set as 3 hours and 0.5 hour respectively. The loading condition is modeled by three load levels with loading factors equal to 70% (2,000 h/year), 83% (5,760 h/year) and 100% (1,000 h/year) of the corresponding peak demand, respectively. Investment decisions are made over a 10-year planning horizon divided into 5 stages whose durations are equally two years. The interest rate is set as 10%, and the load demand data come from [28], and the unit reliability cost of EENS, ω , is set as 10 \$/MWh. The lifespan of conductors and transformers are set as 20 years [45]. The reliability indices of planned distribution network are further checked by the reliability assessment method proposed in [27]. The simulations were implemented on a PC with an Intel Core i7-7700 processor at 3.60 GHz and 32 GB of RAM using CPLEX 12.5. The AC power flows are calculated of the planned distribution networks by MATPOWER [52] and the feasibility of nodal voltage constraints and capacity constraints of equipment are verified.

Two cases are firstly studied in this section. In the first case, we set reliability requirement be relatively loose. The performance of the proposed approach has been compared with the planning model based on the reliability assessment model presented in [28], in which post-fault load restoration through tie-lines is

Fig. 6. Investment decisions of proposed model with $\epsilon_{SAIDI}^t = 3$.Fig. 7. Investment decisions of model based on [27] with $\epsilon_{SAIDI}^t = 3$.

disregarded. In the second case, we set reliability requirement be more strict and different reliability requirement is applied to different load demand. In this case, the model based on [28] becomes infeasible because the radial-constructed network cannot meet the strict reliability requirement. Then, for the proposed model, the sensitivity analyses on the reliability requirement to the total planning cost is further studied. In the last subsection, discussions about unit reliability cost of EENS in the objective function is presented.

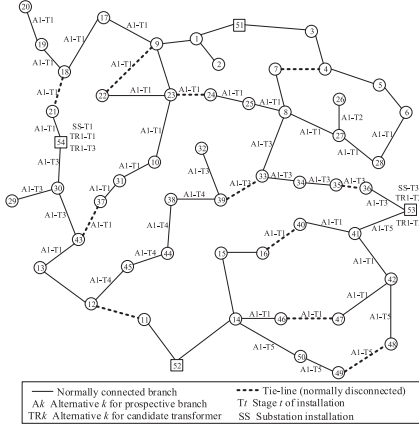
A. Planning Results Under Loose Reliability Requirement

We first set reliability requirement be relatively loose with $\epsilon_{SAIDI}^t = 3$ and $\epsilon_i^{CID,t} = 3, \forall i \in \Psi_t^{LN}$ for all planning stages. The planning results of the proposed model and the planning model based on [28] are depicted in Figs. 6 and 7 respectively.

In Fig. 6, three tie-lines are constructed between nodes 24 and 25, nodes 37 and 43, as well as nodes 42 and 48 respectively, making the network become a mesh network. It can be verified that through post-fault load restoration, demand of nodes which are downstream the outage branch could be effectively transferred to another feeder through tie-lines. In Fig. 7, one more substation and two more transformers at node 53 are to be built by the planning model based on [28] in order to satisfy

TABLE II
RESULTS COMPARISONS

	Proposed Model	Model based on [28]
Total cost (PV) (10⁵\$)	4.827	17.245
SAIDI (h)	1.5128	1.7066
ASAI (%)	99.983	99.981
SAIFI (1/year)	1.7600	1.1677
EENS (MWh/year)	17.872	20.162
Computing time (min.)	49.24	15.66

Fig. 8. Investment decisions of the proposed model with $\epsilon_{SAIDI}^t = 0.867$.

the reliability requirement. In Fig. 7, the network is constructed radially, where three independent parts are separated from each other. Detailed data including planning cost (present value, PV), reliability indices and computing time are listed in Table II.

From Table II we can see that, under the same reliability requirement, the planning results generated by the proposed model are much more economical (which costs about a quarter of the cost based on the model of [28]). Under lower budget, owing to the mesh structure and network reconfiguration strategies, the planned network in Fig. 6 reaches similar reliability indices with the other costly one in Fig. 7. Thus, it is important to take the planning of tie lines into consideration of the reliability assessment and planning models.

B. Planning Results Under Strict Reliability Requirement

In this subsection, we further study a more practical benchmark, where the reliability requirement becomes much stricter and varies node by node, as listed in Table IV. Annual CIDs for levels A+, A, B [40] are set as 5 minutes (0.0833 hour), 52 minutes (0.8667 hour) and 3 hours for last stage respectively, and their corresponding nodal ASAI are 99.999%, 99.990% and 99.966%. The SAIDIs for whole network during all stages are set as 52 minutes for an A-level network. The final planning results are depicted in Fig. 8. The total cost (present value) reaches 35.457 (10⁵\$), and the SAIDI is equal to 0.856 hour. The model is solved in 52.75 minutes.

Detailed results for each planning stage are listed in Table III. As depicted in Fig. 8, two new substations, 4 new transformers and 45 new branches are built and eleven tie-lines exist at the last planning stage. For the nodes leveled 'A+', the length of

TABLE III
RESULTS OF EACH PLANNING STAGE

Planning Stage	1 st	2 nd	3 rd	4 th	5 th
Number of new subs.	1	0	1	0	0
Number of new trans.	1	0	2	0	1
Number of new branches	25	1	10	4	5
Cost of Stage (PV) (10⁵\$)	15.756	0.519	14.692	1.791	2.699
Cost of EENS (\$)	58.35	76.12	94.54	122.14	146.85
SAIDI (h)	0.8631	0.8661	0.8217	0.8338	0.8562
SAIFI (1/year)	0.8933	1.0342	0.7721	0.8124	0.7971
EENS (MWh/year)	5.835	7.612	9.454	12.214	14.685

TABLE IV
NODAL CIDS AT THE LAST PLANNING STAGE

Node	Level	CID(h)	Node	Level	CID(h)	Node	Level	CID(h)
1	A	0.709	18	B	1.149	35	A	0.819
2	B	0.874	19	B	1.278	36	A+	0.064
3	A	0.819	20	B	1.518	37	A	0.760
4	A	0.819	21	A+	0.029	38	B	2.635
5	A	0.819	22	A	0.709	39	B	2.771
6	A	0.819	23	A	0.709	40	A	0.458
7	A	0.819	24	A	0.819	41	A	0.547
8	A	0.819	25	A	0.819	42	A	0.688
9	B	1.917	26	B	1.788	43	A	0.658
10	A	0.709	27	A	0.819	44	A	0.635
11	A+	0.075	28	A	0.819	45	A	0.635
12	A	0.635	29	B	1.309	46	A	0.471
13	A	0.635	30	A	0.635	47	A	0.412
14	A	0.471	31	A	0.709	48	A	0.688
15	A	0.471	32	B	2.885	49	A	0.747
16	A	0.471	33	B	0.819	50	A	0.471
17	B	1.564	34	A	0.819			

supplying feeders is short. For example, for nodes 11, 21 and 36 whose reliability level are "A+", the feeders only supply these nodes without any node downstream. Nodal CIDs are listed in Table IV. With the nodal reliability requirement constraints (53)–(54), precise designs for the planning of distribution networks can be achieved, and the total planning cost is minimized while meeting the reliability requirement.

C. Sensitivity Analyses of the Reliability Requirement

In this subsection, the sensitivity analyses highlighting the costs/benefits of different reliability requirement are carried out. Only the requirement of SAIDI varies from case to case for convenience, and the nodal reliability requirement listed in Table III is neglected here. The final planning results of the 54-node networks under different requirement, $\epsilon_{SAIDI}^t, t = 1, 2, \dots, T$, are depicted as Fig. 9, Fig. 10 and listed in Table V.

As shown in Fig. 9, as the requirement of SAIDI decreases, the number of feeders and tie lines increases. As shown in Table V and Fig. 10, the total planning cost firstly grows slowly when the reliability requirement is relatively loose and then rises rapidly when the reliability requirement of SAIDI further decreases. As listed in Table V, the SAIDIs of the planned distribution network are closely less than the reliability requirement of SAIDI, ϵ_{SAIDI}^t . Therefore, with the proposed reliability-constrained planning model of distribution networks, planners can economically and precisely obtain several planning schemes with different reliability requirement.

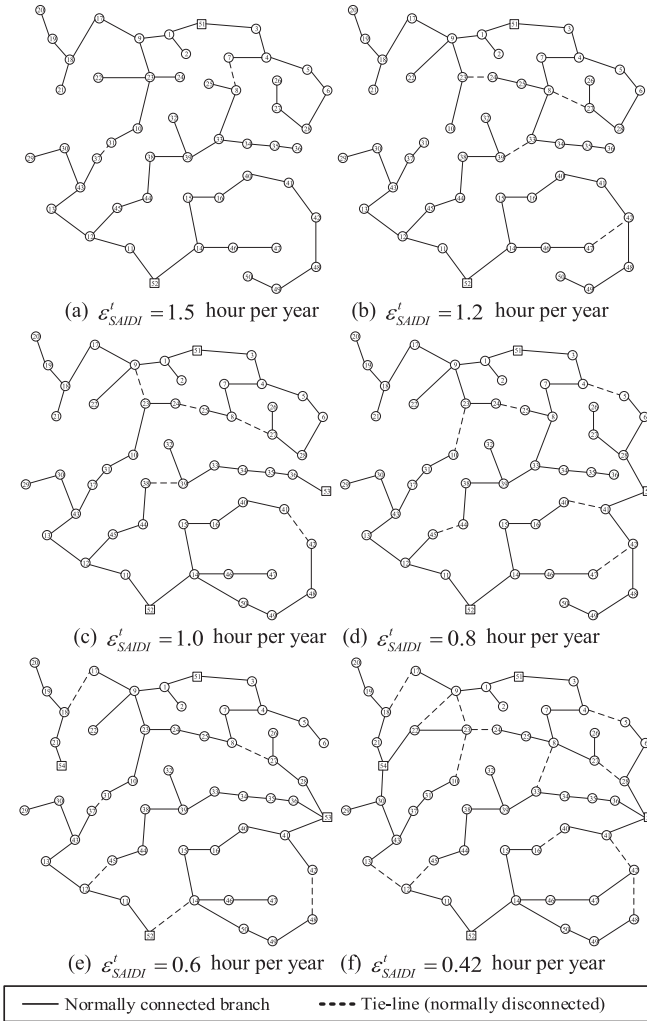


Fig. 9. Planning results (at the last stage) for different reliability requirement.

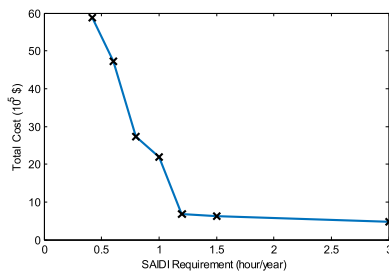


Fig. 10. Total planning cost under different reliability requirement.

D. Sensitivity Analyses of the Unit Reliability Cost of EENS in the Objective Function

In this subsection, the sensitivity analyses of the unit reliability cost of EENS, ω , in the objective function are further discussed. The requirement of SAIDI is set as 1.5 hour/year and the nodal reliability requirement is also neglected here. The results are listed in Table VI.

As shown in Table VI, with the increase of ω , the part of the reliability cost in the objective function weights more, and the

TABLE V
RESULTS UNDER DIFFERENT RELIABILITY REQUIREMENT

SAIDI requirement (h/year)	Total Cost (PV) (10 ⁵ \$)	Reliability indices for the last planning stage		
		SAIDI (h/year)	SAIFI (1/year)	EENS (MWh/year)
1.5	6.227	1.4470	1.6689	17.094
1.2	6.771	1.1834	1.5672	13.980
1.0	21.878	0.9301	1.3385	10.987
0.8	27.373	0.7885	1.1617	9.315
0.6	47.238	0.5832	1.0310	6.890
0.42	58.748	0.4154	0.7851	4.908

TABLE VI
PLANNING RESULTS UNDER DIFFERENT UNIT RELIABILITY COST OF EENS

ω (\$/MWh)	Total Cost (10 ⁵ \$)	Investment and Maintenance Cost (10 ⁵ \$)	Reliability Cost (10 ⁵ \$)	Reliability indices for the last planning stage		
				SAIDI (h/year)	SAIFI (1/year)	EENS (MWh/year)
10	6.227	6.226	0.002	1.4470	1.6689	17.094
100	6.435	6.418	0.017	1.4282	1.5799	16.872
1000	6.610	6.442	0.168	1.4236	1.6087	16.818
10000	8.127	6.483	1.644	1.3916	1.6836	16.440
100000	19.627	6.939	12.688	1.0740	1.5746	12.688

SAIDI and EENS gradually decrease. However, an overlarge unit reliability cost of EENS will make the reliability constraint of SAIDI become “invalid”, which can be seen from Table VI that the SAIDI is much smaller than the SAIDI requirement when $\omega = 100\,000$. Thus, a relatively small unit reliability cost of EENS is recommended in the objective function.

V. CONCLUSION

In this paper, a reliability-constrained expansion planning model for mesh distribution networks is presented. Post-fault load restoration between feeders is fully considered, providing probability of reducing investment cost under same predefined reliability requirement for the meshed-constructed networks than the radial-constructed networks. Reliability requirement is guaranteed precisely according to the reliability constraints, which can generate accurately economic planning schemes for planners. Systematical and nodal reliability requirement can be applied individually based on the importance of the demands, which can be more flexible in practical planning projects. The degradation of power equipment is also considered in the present model. Last but not the least, the reliability-constrained planning model is formulated as an instance of mixed-integer linear programming and can be effectively solved by off-the-shelf commercial software without heuristic iterations or tentative simulations, which is more effective and efficient. For future work, as a type of power injection to the distribution network, distributed generation can be further taken into account by utilizing multiple-scenario method.

TABLE VII
ERRORS INTRODUCED BY POWER FLOW LINEARIZATION

Terms		Relative error
Nodal voltages	Max.	0.051
	Ave.	0.012
Active power of branches	Max.	0.074
	Ave.	0.033
Reactive power of branches	Max.	0.021
	Ave.	0.014

APPENDIX

Based on the distflow model proposed in [46], the power flow through branch ij can be expressed as

$$U_i - U_j = 2(r_{ij}P_{ij} + x_{ij}Q_{ij}) - (r_{ij}^2 + x_{ij}^2) \frac{P_{ij}^2 + Q_{ij}^2}{U_i} \quad (69)$$

where U_i is the square of the voltage at node i . Notice that the nonconvex term $(r_{ij}^2 + x_{ij}^2) \frac{P_{ij}^2 + Q_{ij}^2}{U_i}$ represents the power loss. In order to optimally solve the proposed planning model by optimization solvers, we abandon this term for relative small value and the equation (69) is then simplified as

$$U_i - U_j = 2(r_{ij}P_{ij} + x_{ij}Q_{ij}) \quad (70)$$

For the case studies demonstrated in Section IV based on the 54-node system, posterior check of the inaccuracies introduced by the linearization is carried out for the network in Fig. 6, as listed in Table VII. The AC power flow is calculated by MATPOWER [52].

From Table VII it can be inferred that the relative errors of nodal voltages and power flows are acceptable for the planning model. If we define the loss ratio lr as

$$lr = \frac{(r_{ij}^2 + x_{ij}^2) \frac{P_{ij}^2 + Q_{ij}^2}{U_i}}{2|r_{ij}P_{ij} + x_{ij}Q_{ij}|} \quad (71)$$

based on the results calculated by MATPOWER, the maximum loss ratio and average loss ratio are 0.076 and 0.044 respectively, which makes it possible to abandon the power loss term in (69).

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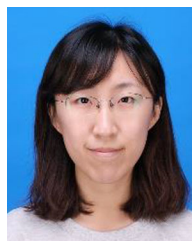
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